Theoretical Exercises – Sheet 8 - Solution

1. How is Data Science different from Data Mining? How are they similar?

   Data Mining is a process in which we extract knowledge from raw data. We aim to discover interesting patterns and gain knowledge by applying algorithms. Data Mining is a crucial step in the process of Knowledge Discovery in Databases (KDD).

   Data Science, on the other hand, is a broader field that encompasses the entire lifecycle of working with data. It involves collecting, cleaning, transforming, analyzing, and visualizing data to derive insights and make data-driven decisions. While Data Mining focuses specifically on extracting patterns and knowledge from data, Data Science integrates multiple disciplines such as statistics, machine learning, data engineering, and domain expertise to solve complex problems. Essentially, Data Mining is a subset of the larger field of Data Science.

2. How does Data Mining relate to Machine Learning?

   Machine Learning is closely related to Data Mining. Data Mining involves the systematic application of statistical and algorithmic methods to large datasets to uncover new patterns, correlations, and trends. Many of the algorithms used in ML are also applied in Data Mining. The knowledge gathered from Data Mining can be further used to prepare data for Machine Learning models or to generate training datasets. Data Mining is, therefore, an important interface that both utilizes Machine learning algorithms and supports their results.

3. Go through the CRISP-DM model and describe in your own words what happens in each phase.

   Business Understanding: Definition of goals and requirements; Derivation of the concrete task and the rough procedure

   Data Understanding: Data collection or first inspection of the available data; Identify potential data quality issues

   Data Preparation: Cleaning raw data, transforming to the according format, preparing for modeling.

   Modeling: Application of suitable data mining methods, optimization of parameters; usually determination of several models and choosing correct algorithms

   Evaluation: Selection of the model that best fulfills the task. Careful comparison with the task.
   Checking if model accuracy is sufficient and if model performs well.

   Deployment: Processing and presentation of the results; possible integration of the model into a decision-making process of the client

4. Present an example where Data Mining is crucial to the success of a business. What data mining functionalities does this business need? Formulate corresponding business questions.

   Data mining is a process used by companies to turn raw data into useful information. By using software to look for patterns in large batches of data, businesses can learn more about their customers to develop more effective marketing strategies, increase sales and decrease costs. Moreover, Data mining can help spot sales trends, develop smarter marketing campaigns, and accurately predict customer loyalty.

   Association: By analyzing shopping patterns, Walmart discovered that young fathers who bought diapers often also purchased beer during late-night store visits. To take advantage of this insight, the store placed diapers and beer next to each other. As a result, sales of both products increased significantly.
   Source:
   https://www.forbes.com/forbes/1998/0406/6107128a.html

Clustering: An example where data mining is crucial to business success is Netflix's use of clustering to analyze binge-watching behavior. By applying cluster analysis, similar to the study on binge-watching behaviors, Netflix segments users into different groups, such as "Recreational Watchers" (casual viewers), "Regulated Binge-Watchers" (those who watch in moderation), "Avid Binge-Watchers" (highly engaged but controlled), and "Unregulated Binge-Watchers" (who may exhibit problematic watching patterns).
Source:
https://www.researchgate.net/publication/335589973_Overcoming_the_unitary_exploration_of_binge-watching_A_cluster_analytical_approach

Classification: E-Mail services such as outlook or gmail use classification to find out whether a Mail is considered as spam or not.

Source:
https://www.researchgate.net/publication/46122402_An_Efficient_Two-phase_Spam_Filtering_Method_Based_on_E-mails_Categorization

Regression: A retail company can use multiple regression to forecast how variables like holiday seasons, advertising budget, and competitor pricing impact future revenue.

5. Explain the difference between Association, Segmentation/Clustering, Classification and Regression.

Association: Is about finding patterns or relationships between items in a dataset, often used in market basket analysis. It identifies which items are frequently purchased together. The purpose of it is to uncover hidden patterns and rules. An example Business Question might be "What items should be displayed together in a store?"

Clustering: Divides data into groups based on similarities, without predefined labels. It helps to understand different groups or patterns within a dataset. An Example business question might be “Who are my web visitors and how can we target certain groups for advertising.

Classification: Predicts categories or labels for data points based on previous training. It is used for supervised learning, where data is labeled. We want to classify new instances into given categories. Business questions could be “Is this email spam or not?”.

Regression: Predicts a continuous numerical value based on historical data and independent variables. It is used for understanding relationships and making predictions. With regression we want to predict quantitative outcomes. Example for a Business question: “What will the sales of ice cream be next month?”

6. Have a look at the efforts in Data Science per phase. Why do you think is data preparation so much effort and why is evaluation not?

Data preparation is high effort since we must make sure that our data is has a uniform format. Since we might get our Data from different sources this might lead different formats. Processing large datasets might require a lot of resources specially if someone must correct them manually.

Evaluation: Evaluation requires less effort because we can use modern tools and frameworks which give us evaluation metrics. Making the whole process faster and more straight forward. Once we have our Data prepared and our insights/models are trained. Evaluating the model is often a structured and automated Task.

All in all, we can say that data preparation takes more effort because cleaning data takes more time and dealing with messy and unstructured data takes a lot of resources, while evaluation is more straightforward since we can use automated tools and predefined metrics to assess the performance of the models

7. Provide examples for bad data quality.

Examples for bad data quality:

Missing Data: Dataset with missing values. Could lead to biased results or incomplete analysis.

Inconsistent Data: Dataset where multiple formats are being used for the same data. E.g. Data formats are inconsistent, one format is 01/02/2025 the other is 2025/02/01

Duplicate Data: Multiple Entries could lead to biased results.

Incorrect Data: Incorrect Data E.g. Price of a product might be listed as -100 CHF instead of 100 CHF

8. Provide examples for each attribute type: nominal, ordinal, interval-scaled and ratio-scaled and decide whether it makes sense to calculate mean, median and/or mode.

Nominal:
- Examples: presence of something, Colors, Citys, Countries, Company Names
- Can be coded, but mathematical operations do not make sense
- Mean and Median cannot be computed, only the mode

Ordinal:
- Examples: Grades, Sizes such as S, M, L, XL
- Can be coded, but mathematical operations do not make sense
- Mode and (maybe) Median gives a tendency, Mean does not make sense

Cardinal:
- Examples: Temperature, IQ, Age, Salaries, Weight, Height
- Mode, Median and Mean makes sense. Mode can be complicated without binning

Interval-Scaled (Cardinal):
- Examples: Temperature, Calender Year
- Mode, Median and Mean makes sense. Mode can be complicated without binning

Ratio-Scaled (Cardinal):
- Examples: Weight, Height, Income
- Mode, Median and Mean makes sense. Mode can be complicated without binning

Note that the difference between Ratio-scaled Attributes and Interval scaled attributes rely on the absolute zero point. Since we can say that having the weight of 0 means that there is no weight. On the other hand, we cannot say that the year 0 is the beginning of everything.

9. Suppose that the data for analysis includes the attribute *age*. The *age* values for the data tuples are 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70 (in increasing order).

   a. What is the mean of the data? What is the median?

      N = 27
      Mean = 809/27 = 29.963
      Median = 25

   b. What is the mode of the data? Comment on the data's modality.

      Mode = {25, 35} – Bimodal / Multimodal

   c. Can you find (roughly, by hand) the first quartile Q_1 and the third quartile Q_3 of the data?

      Median divides into 2 subsets (each again with the median) à 14 values.
      Q1 is between the 7th and 8th value (20 and 21).
      Quartile Q1 = 20.5

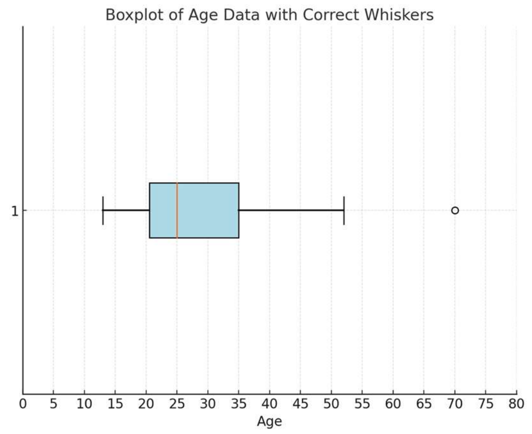
Quartile Q3 = 35

Extra:
Range = 57
IQR = 14.5

d. Give the five-number summary of the data and show a boxplot.

Minimum = 13
Quartile Q1 = 20.5
Median = 25
Quartile Q3 = 35
Maximum = 70

Whiskers from 13 to 52, 70 outside range 1,5 x IQR



10. Suppose that a hospital tested the age and the body fat data for 18 randomly selected adults as follows:

age	23	23	27	27	39	41	47	49	50
%fat	9.5	26.5	7.8	17.8	31.4	25.9	27.4	27.2	31.2

age	52	54	54	56	57	58	58	60	61
%fat	34.6	42.5	28.8	33.4	30.2	34.1	32.9	41.2	35.7

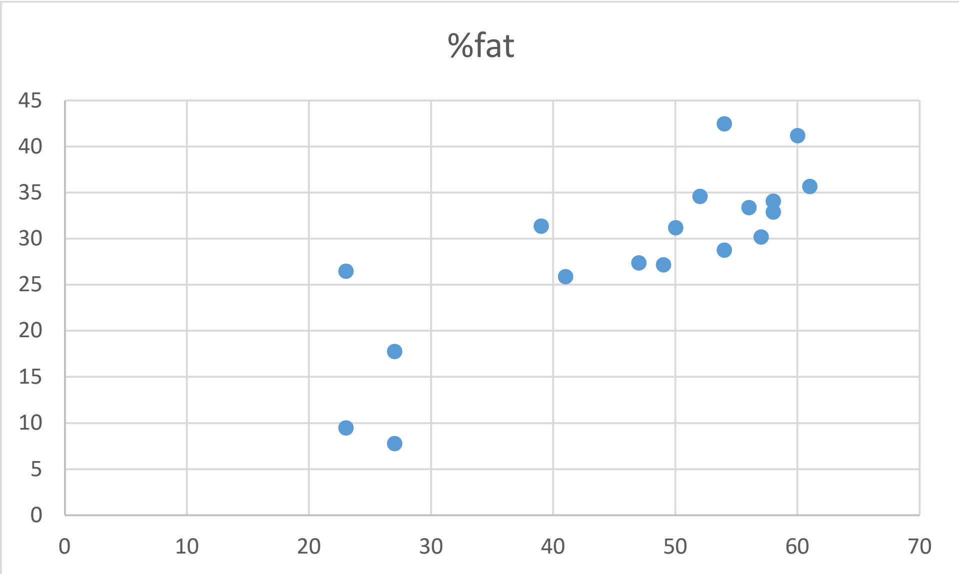
a. Calculate the mean, median and standard deviation of *age* and *%fat*.

For age:
Mean = 46.4
Minimum = 23
Quartile Q1 = 39
Median = 51
Quartile Q3 = 57
Maximum = 61
Variance = 165.0247
Standard Deviation = 12.846

For %fat:
Mean = 28.783
Minimum = 7.8
Quartile Q1 = 26.5
Median = 30.7 – wenn nicht sortiert, dann kommt man aus 32.9 (typischer Fehler!)
Quartile Q3 = 34.1
Maximum = 42.5

Variance = 80.886
Standard Deviation = 8.994

b. Draw a scatter plot based on these two attributes.



We can see that there is a strong correlation between Age and Fat%. Calculating the Correlation Coefficient, we get a value of 0.82 which confirms our Statement. Note that a close value to 1 means that we have a strong positive correlation while having a correlation of -1 indicates having a strong negative correlation.